THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 02

Full Syllabus (Chapters 1–13) | Class 8 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** a smaller lit part is visible from Earth	Phases depend on relative positions
2	**(a)** a testable expectation about what will happen	Tested by experiment
3	**(b)** methane	Burns to give heat
4	**(c)** bacterial infections	They target bacterial cells
5	**(c)** body, mind and surroundings	Dinacharya and ritucharya
6	**(b)** lemon juice	Acidic liquid
7	**(b)** they contain harmful metals	Lead, cadmium, nickel, lithium
8	**(c)** greater than buoyant force	Net force downward
9	**(c)** Kerala, Karnataka and Tamil Nadu	Help mangoes ripen
10	**(d)** lower air pressure	Bernoulli-type effect
11	**(a)** shape and volume	Strong forces, no free movement
12	**(c)** 4 : 1	Tamra 4 parts and Vanga 1 part
13	**(b)** compound	New substance formed
14	**(d)** it insulates water below for aquatic life	Fish survive in cold weather
15	**(b)** diverge	Diverging lens
16	**(c)** ancient Indian astronomical text	Mentions Makar and solstices
17	**(a)** parasitism	Parasite benefits, host harmed
18	**(d)** carnivores	Flesh-eaters
19	**(a)** after the egg is laid	During hatching
20	**(a)** Reuse, repair and recycle	Reduce pollution

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Viruses reproduce only inside host cells (1). Bacteria are cells that can live and reproduce on their own (1).	1+1
22	Use antibiotics only when prescribed (1). Take the correct dose for the full duration (1).	1+1
23	Galvani thought electricity came from the frog (1). Volta showed it came from metals and liquid (1).	1+1
24	Charges separate in clouds by rubbing of ice and water (1). They discharge suddenly when air insulation breaks (1).	1+1
25	Natural solid substances of fixed chemical composition (1). Mostly compounds, rarely elements (1).	1+1
26	Gas is less soluble at higher temperature (1). Gas escapes (1).	1+1
27	National parks (1). Wildlife sanctuaries or biosphere reserves (1).	1+1
28	Pollination: transfer of pollen to stigma (1). Fertilisation: fusion of male and female gametes (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Grade 6: wonder (1). Grade 7: science evolves (1). Grade 8: investigation combines both (1).	1+1+1
30	No, it acts without contact (1½). Earth pulls objects falling towards it even from a distance (1½).	1½+1½
31	Yes as ice, water, vapour (1). Same particles in different arrangements (1). Depends on temperature (1).	1+1+1
32	Convex mirror (1½). Concave lens (1½).	1½+1½
33	Rotation on axis defines a day (1½). Revolution round the Sun defines a year (1½).	1½+1½
34	Species disappear due to habitat loss (1½). Ecosystems become weaker (1½).	1½+1½
35	Moving liquid iron in the core forms currents (1). Currents produce Earth's magnetic field (1). Animals use it to navigate (1).	1+1+1

	OR: Dry cell is single-use (1). Rechargeable can be reused many times (1). Rechargeable wears out after many cycles (1).	1+1+1
36	They depend on the ecosystem (1½). Their knowledge helps manage it (1½).	1½+1½
	OR: Provide conditions to live (1½). Examples: sunlight, water, soil, temperature (1½).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	Mould on a dish stopped bacteria (1). Penicillin was the first antibiotic (1). Antibiotics treat bacterial infections (1). Vaccines prevent diseases by training immunity (1). Antibiotics do not work on viruses (1).	1+1+1+1+1
	OR: Both build memory (1). Infection involves actual illness (1). Vaccination is safe and controlled (1). Vaccination protects the community (1). Example: chickenpox (1).	1+1+1+1+1
38	0.1 m^2: 400 N/m^2 (1½). 0.05 m^2: 800 N/m^2 (1½). Smaller area means more pressure (2).	1½+1½+2
	OR: Kalboishakhi in West Bengal, Bihar, Jharkhand (1½). Bordoila in Assam (1). Mango showers in southern states (1). Help kharif crops, mangoes and coffee (1½).	1½+1+1+1½
39	Solubility decreases (1). Excess solute separates (1). Crystals form (1). Example – sugar or salt crystals (1). Slow cooling gives bigger crystals (1).	1+1+1+1+1
	OR: Sea water has salts (1). Blood carries dissolved substances (1). Oxygen dissolved for fish (1). Alloys in industry (1). Medicines as solutions (1).	1+1+1+1+1
40	National parks (1). Sanctuaries (1). Biosphere reserves (1). Community areas (1). Importance (1).	1+1+1+1+1
	OR: (i) Algae → small fish → big fish (1). (ii) Producer algae, consumers fish (1). (iii) Decomposers recycle (1). (iv) Big fish suffer (1). Algae increase (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER